Kopecka–Muller and the Nonperiodic Projection Strong-Convergence Question

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2026-07-18

Status

Literature-implied answer. This is not a new proof. The later literature gives a negative answer to the source paper's nonperiodic strong-convergence question after identifying it with the Amemiya–Ando norm-convergence problem for products of orthogonal projections.

Source Question

The source paper is P. L. Combettes and N. N. Reyes, *Functions with Prescribed Best Linear Approximations*, arXiv:0905.3520. In Section 3, p. 11, after the periodic projection algorithm (3.8) and the affine projectors (3.9), the authors note that for nonperiodic activation of the projectors Amemiya–Ando established weak convergence, and ask whether "strong convergence holds."

general not of direct numerical use since it requires the inversion of operators in (3.1). However, minimal norm solutions and, more generally, best approximations from the solution set of (1.1) can be computed iteratively via projection methods. Indeed, for every $r \in \mathcal{H}$ and $(u_i)_{1\leq i\leq m} \in \times_{i=1}^m U_i$, let us denote by $B(r; u_1, \dots, u_m)$ the best approximation to r from $S(u_1, \dots, u_m)$, i.e., by Proposition 2.1(i),

$$B(r; u_1, \dots, u_m) = P_{S(u_1,\dots,u_m)} r = P_{\bigcap_{i=1}^m (u_i + U_i^\perp)} r. \tag{3.7}$$

A standard numerical method for computing $B(r; u_1, \dots, u_m)$ is the periodic projection algorithm

$$x_0 = r \quad \text{and} \quad (\forall n \in \mathbb{N}) \quad x_{n+1} = Q_1 \cdots Q_m x_n \tag{3.8}$$

where, for every $i \in \{1, \dots, m\}$, Q_i is the projector onto $u_i + U_i^\perp$, i.e.,

$$Q_i = P_{u_i + U_i^\perp} \colon x \mapsto u_i + x - P_i x. \tag{3.9}$$

This algorithm is rooted in the classical work of Kaczmarz [25] and von Neumann [39]. Although it has been generalized in various directions [3, 4, 9, 15], it is still widely used due to its simplicity and ease of implementation. If $S(u_1, \dots, u_m) \neq \varnothing$, the sequence $(x_n)_{n\in\mathbb{N}}$ generated by (3.8) converges strongly to $B(r; u_1, \dots, u_m)$. If $u_i \equiv 0$, this result was first established by von Neumann [39] for $m = 2$ and extended by Halperin [22] for $m > 2$. Strong convergence to $B(r; u_1, \dots, u_m)$ in the general affine case ($u_i \not\equiv 0$) is a routine modification of Halperin's proof via Lemma 2.9 (see [18] for a detailed account). Interestingly, if the projectors are not activated periodically in (3.8) but in a more chaotic fashion, only weak convergence has been established [1] and it is still an open question whether strong convergence holds.

In connection with (3.8), an important question is whether the convergence of $(x_n)_{n\in\mathbb{N}}$ to $B(r; u_1, \dots, u_m)$ occurs at a linear rate. The answer is negative and it has actually been shown that arbitrarily slow convergence may occur [5] in the sense that, for every sequence $(\alpha_n)_{n\in\mathbb{N}}$ in $]0,1[$ such that $\alpha_n \downarrow 0$, there exits $r \in \mathcal{H}$ such that

$$(\forall n \in \mathbb{N}) \quad \|x_n - B(r; u_1, \dots, u_m)\| \geq \alpha_n. \tag{3.10}$$

On the other hand, several conditions have been found [3, 5, 6, 17, 19, 26] that guarantee that, if (1.1) admits a solution for some $(u_i)_{1\leq i\leq m} \in \times_{i=1}^m U_i$, then, for every $r \in \mathcal{H}$, the sequence $(x_n)_{n\in\mathbb{N}}$ generated by (3.8) converges uniformly linearly to $B(r; u_1, \dots, u_m)$ in the sense that there exists $\alpha \in [0,1[$ such that [19, Section 4]

$$(\forall n \in \mathbb{N}) \quad \|x_n - B(r; u_1, \dots, u_m)\| \leq \alpha^n \|r - B(r; u_1, \dots, u_m)\|. \tag{3.11}$$

The next result states that the IBAP implies uniform linear convergence of the periodic projection algorithm for solving the underlying affine feasibility problem (1.1) for every $(u_i)_{1\leq i\leq m} \in \times_{i=1}^m U_i$ and every $r \in \mathcal{H}$. In other words, if (1.1) admits a solution for every $(u_i)_{1\leq i\leq m} \in \times_{i=1}^m U_i$, then uniform linear convergence always occurs in (3.8).

Proposition 3.2 *Suppose that $(U_i)_{1\leq i\leq m}$ satisfies the inverse best approximation property and set*

$$\alpha = \sqrt{1 - \prod_{i=1}^{m-1} \Big(1 - \mathsf{c}\big(U_i^\perp, U_{i+}^\perp\big)^2\Big)}. \tag{3.12}$$

Then $\alpha \in [0,1[$ and, for every $r \in \mathcal{H}$ and every $(u_i)_{1\leq i\leq m} \in \times_{i=1}^m U_i$, the sequence $(x_n)_{n\in\mathbb{N}}$ generated by (3.8) satisfies (3.11).

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Supporting Answer

The supporting paper is E. Kopecka and V. Muller, *A product of three projections*, Studia Mathematica 223 (2014), 175–186, DOI 10.4064/sm223-2-4. Its abstract identifies the Amemiya–Ando question as the long-open norm convergence problem for arbitrary sequences of orthogonal projections onto a finite family of closed subspaces. It then states that, after Paszkiewicz's five-subspace example, the paper constructs three such subspaces and resolves the problem fully.

STUDIA MATHEMATICA 223 (2) (2014)

A product of three projections

by

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Abstract. Let X and Y be two closed subspaces of a Hilbert space. If we send a point back and forth between them by orthogonal projections, the iterates converge to the projection of the point onto the intersection of X and Y by a theorem of von Neumann.

Any sequence of orthoprojections of a point in a Hilbert space onto a finite family of closed subspaces converges weakly, according to Amemiya and Ando. The problem of norm convergence was open for a long time. Recently Adam Paszkiewicz constructed five subspaces of an infinite-dimensional Hilbert space and a sequence of projections on them which does not converge in norm. We construct three such subspaces, resolving the problem fully. As a corollary we observe that the Lipschitz constant of a certain Whitney-type extension does in general depend on the dimension of the underlying space.

1. Introduction. Let K be a fixed natural number and let $\mathscr{L} = \{L_1, \dots, L_K\}$ be a family of K closed subspaces of a Hilbert space H. Let $z_0 \in H$ and $k_1, k_2, \dots \in \{1, \dots, K\}$ be arbitrary. Consider the sequence of vectors $\{z_n\}$ defined by

$$z_n = P_{k_n} z_{n-1}, \tag{1}$$

where P_k denotes the orthogonal projection of H onto L_k. The sequence $\{z_n\}$ converges weakly by a theorem of Amemiya and Ando [AA]. If each projection appears in the sequence $\{P_{k_n}\}$ infinitely many times, then this limit is equal to the projection of z_0 onto the intersection of all spaces in $\mathscr{L}$.

If $K = 2$ then the sequence $\{z_n\}$ converges even in norm according to a classical result of von Neumann [N].

If $K \geq 3$ then additional assumptions are needed to ensure the norm-convergence. That $\{z_n\}$ converges if H is finite-dimensional was originally proved by Práger [Pr]; this also follows, of course, from [AA].

If H is infinite-dimensional, but the sequence $\{k_n\}$ is periodic, the sequence $\{z_n\}$ converges in norm according to Halperin [Ha]. The result was generalized to quasiperiodic sequences by Sakai [S]. Recall that the sequence

2010 *Mathematics Subject Classification*: Primary 46C05; Secondary 54C20.
Key words and phrases: Hilbert space, projection, product, extension.

DOI: 10.4064/sm223-2-4

Identification

In a Hilbert space, strong convergence of a vector sequence is exactly norm convergence. The source's affine projection setup contains the homogeneous linear subspace case by taking the affine shifts to be zero. Thus a sequence of orthogonal projections onto finitely many closed linear subspaces that fails to converge in norm is already a counterexample to the universal strong convergence assertion for nonperiodic activation.

Kopecka–Muller supply such an example using three closed subspaces of an infinite-dimensional Hilbert space. Therefore the source's question has a negative answer in the linear subspace subcase, and hence a negative answer as stated for arbitrary nonperiodic activation.

Scope Limitations

This packet does not concern the periodic projection algorithm, the source's IBAP characterizations, or the source's later theorem that IBAP implies uniform linear convergence for the periodic algorithm. It only clears the nonperiodic/chaotic activation strong-convergence signal. The supporting authors explicitly resolve the Amemiya–Ando norm-convergence problem; the connection to this particular arXiv:0905.3520 sentence is an agent-identified implication rather than a direct citation to Combettes–Reyes.

Search Evidence

Local run indexes were searched for arXiv:0905.3520, the source title, prescribed best linear approximations, Amemiya–Ando, Kopecka–Muller, Paszkiewicz, projection strong convergence, and projection norm convergence. No exact prior packet for this source question was found. Web checks on 2026-07-18 found the arXiv source metadata and EUDML/IMPAN metadata for the Kopecka–Muller paper, including the abstract statement that three subspaces resolve the norm-convergence problem fully.

References

[1] P. L. Combettes and N. N. Reyes. *Functions with Prescribed Best Linear Approximations*. arXiv:0905.3520, 2009.

[2] I. Amemiya and T. Ando. *Convergence of random products of contractions in Hilbert space*. Acta Sci. Math. (Szeged) 26 (1965), 239–244.

[3] E. Kopecka and V. Muller. *A product of three projections*. Studia Mathematica 223 (2014), 175–186. DOI: 10.4064/sm223-2-4.